

CSES Introductory Problems

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This is a Typst adaptation of my website. Watch out for hyperlinks; a lot of things are clickable.

Problem 1

Consider an algorithm that takes as input a positive integer n . If n is even, the algorithm divides it by two, and if n is odd, the algorithm multiplies it by three and adds one. The algorithm repeats this, until n is one. For example, the sequence for $n = 3$ is as follows:

$$3 \rightarrow 10 \rightarrow 5 \rightarrow 16 \rightarrow 8 \rightarrow 4 \rightarrow 2 \rightarrow 1$$

Your task is to simulate the execution of the algorithm for a given value of n .

This is fairly straightforward. We will just do as we are told:

Solution 1

```
1 while  $n > 1$ 
2   if  $n$  is odd
3      $\lfloor n \leftarrow \frac{n}{2}$ 
4   else
5      $\lfloor n \leftarrow 3n + 1$ 
```

Problem 2

You are given all numbers between $1, 2, \dots, n$ except one. Your task is to find the missing number.

My first thought was to loop through the list, storing which values we encounter. But we can do better! Sum the list and compare it with $\frac{n(n+1)}{2}$.

Solution 2

```
1  $s \leftarrow 0$ 
2 for  $i$  in  $nums$ 
3    $\lfloor s \leftarrow s + i$ 
4 return  $\frac{n(n+1)}{2} - s$ 
```

Problem 3

You are given a DNA sequence: a string consisting of characters A, C, G, and T. Your task is to find the longest repetition in the sequence. This is a maximum-length substring containing only one type of character.

This is asking us to find the length of the maximal contiguous subsequence. Since only the length is asked, we can simply loop through the list, count the length of each block of letters, and update if necessary.

Solution 3

```
1 max_length ← 0
2 temp ← 0
3 for  $i$  in  $len(string)$ 
4   if string()
5 return  $\frac{n(n+1)}{2} - s$ 
```
